

Project Technical Baseline Specification

Multi Sector Technology Park Spain

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1 Introduction and Scope

This document establishes the critical design input parameters and system architectural specifications required for the professional engineering documentation of a multi sector technology park in Spain. All design elements are predicated on compliance with the **Reglamento Electrotécnico para Baja Tensión (REBT)** and relevant Medium Voltage (MT) regulations.

The park aggregates diverse loads, including administrative offices, a public access museum, heavy industrial facilities, and a critical 1 MW nuclear research reactor. The combined approximate demand necessitates a primary Medium Voltage (MT) intake.

2 Architectural and Demand Parameters

These parameters define the physical scale and power requirements for the design baseline. Note that P connected denotes the estimated connected power.

Table 1: **Facility Power and Classification Baseline**

Facility / Sector	REBT Class (ITC)	Area (m ²)	P conn. (kW)	cos ϕ	Supply (V)
Office Building	ITC BT 28 (Oficinas)	1,500	100	0.95	400/230
Museum	ITC BT 28 (Publ. Conc.)	2,500	180	0.90	400
Machining Workshop	ITC BT 29 (Industrial)	1,000	350	0.85	MT → 400
Chemical Plant	ITC BT 29 (Riesgo Especial)	3,000	650	0.88	MT → 400
Nuclear Reactor	Critical/Special Load	500 (Aux.)	1000	0.98	Primary MT
TOTAL PARK DEMAND			≈ 2,300		

3 Primary Distribution and Substation Architecture

The approximate 2.3 MW aggregated demand requires a primary Medium Voltage (MT) intake.

3.1 Medium Voltage (MT) System Parameters

- **Service Voltage (U MT):** 20 kV. (Standard voltage for industrial parks in Spain).
- **Primary MT Intake (Acometida):** Single connection point subject to DNO (Distribution Network Operator) technical standards.
- **Internal MT Network:** Underground network linking the intake to the internal Transformation Centers (CT).

3.2 Transformation Centers (CT) Specification

Three CTs are mandated to manage demand and minimize Low Voltage (BT) line losses. The BT side of all internal CTs will use the **TN S** earthing system (ITC BT 08).

Table 2: Transformation Center (CT) Architecture

CT	Served Load	Primary U	Secondary U	Est. S nominal	BT Earthing
CT A	Workshop and Chem. Plant	20 kV	400 V	1,600 kVA	**TN S**
CT B	Offices and Museum	20 kV	400/230 V	500 kVA	**TN S**
CT C	Nuclear Reactor Aux.	20 kV	400 V	1,000 kVA (Dedicated)	**TN S**

4 Critical Missing Design Parameters

These parameters are essential for safety, stability, and compliance.

- **Short Circuit Power (S cc):** Must be calculated at all busbars to determine the required **breaking capacity** of protective devices (ITC BT 22).
- **Availability/Redundancy Class (Nuclear Reactor):** The critical nature of the reactor requires a Tier IV or similar classification. CT C must have **Dual MT Feeder Redundancy** and a dedicated ****Emergency Power System**** (UPS/Genset).
- **Harmonics Study:** Required due to industrial motor and process loads. Compliance with **EN 50160** necessitates a study to determine the placement and rating of ****Harmonic Filters**** for power quality maintenance.
- **Earthing System (MT Regimen de Neutro):** The MT system's earthing (e.g., Resistor Earthed or Isolated) must be specified by the DNO to ensure protection selectivity and service continuity.
- **EMC Shielding Requirements:** The sensitive equipment in the Chemical Plant and Nuclear Reactor mandates detailed **Electromagnetic Compatibility (EMC) studies** (ITC BT 29) and potentially shielded cabling/conduits.